SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data intergration,Data transformation and data Reduction ,for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks:50**

Annexure A
SENARIO BASED QUESTIONS

Criteria	Understanding of Scenario ***(2)***	Identification of Key Issues ***(2)***	Application of Concepts ***(2.5)***	Decision Making ***(2)***	Clarity of Response ***(1.5)***	***Total(10)***
Weightage	*20%*	20%	25%	20%	15%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

*I kambhoji Thanuja 192373012*_____*(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.
K Thanuja
Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Annexure A
Scenario based Questions

Q1. Data Preprocessing in Retail Analytics *(L3, L4)*

A **retail analytics company** is preparing customer purchase data for building a recommendation system. The dataset contains missing values, inconsistent categorical formats, and extreme values due to data collected from multiple stores and manual entries.

Sample Dataset

Customer_ID	Age	Gender	Monthly_Spend (₹K)	Purchase_Frequency	Loyalty_Level
C1	23	M	20	5	Low
C2	NaN	Male	35	8	Medium
C3	31	F	NaN	10	High
C4	105	Female	500	50	High
C5	27	male	25	NaN	Low
C6	40	FEMALE	45	12	Medium
C7	36	M	38	9	Low
C8	42	Female	300	45	High

Tasks

A. Handling Missing Values

i. Identify missing values in **Age, Monthly_Spend, and Purchase_Frequency**
ii. Apply at least **two imputation techniques** (e.g., Median Imputation, k-NN Imputation)
iii. Compute the imputed values and justify the selected method

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B. Outlier Detection and Treatment

i. Detect outliers in **Age and Monthly_Spend** using the **Z-score method**
ii. Identify records containing extreme values (e.g., Age = 105, Spend = 500)
iii. Apply suitable treatment methods (**capping / transformation / removal**) and justify

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C. Categorical Data Standardization

i. Identify inconsistencies in **Gender** (M, Male, male, FEMALE, etc.)
ii. Convert all entries into a standardized format (**Male / Female**)

Q2. Covariance & Data Reduction in E-commerce Analytics *(L4, L5)*

An **e-commerce platform** is building a predictive model to classify high-value customers. The dataset includes several correlated financial and behavioral attributes, leading to redundancy and increased computational cost.

Sample Dataset

Cust_ID	Age	Income (₹K)	Purchase_Value (₹K)	Credit_Score	Orders	Avg_Balance (₹K)	EMI (₹)	Customer_Class
U1	24	40	100	650	15	12	5000	Low
U2	34	60	150	700	20	18	7000	Medium
U3	44	85	210	720	28	25	9000	Medium
U4	52	120	320	800	35	40	15000	High
U5	29	45	110	680	18	14	5500	Low
U6	39	75	180	710	24	22	8500	Medium
U7	56	150	360	820	40	50	18000	High
U8	37	70	160	705	22	20	8000	Medium

Tasks

A. Covariance Analysis

i. Compute covariance between **Income and Purchase_Value** (show steps or partial calculation)
ii. Interpret whether the relationship is **positive or negative**
iii. Explain how covariance helps in identifying feature relationships

B. Feature Selection

i. Identify redundant attributes using **correlation or domain knowledge**
ii. Select an optimal subset of features for modeling
iii. Justify your selection

C. Data Transformation

i. Standardize the dataset (excluding **Cust_ID and Customer_Class**)
ii. Explain why normalization/standardization is required

Q3. Data Transformation & Discretization in Education Analytics *(L3, L4, L5)*

An **education analytics team** is developing a model to predict student performance. Continuous variables such as study hours and test scores need to be discretized to improve interpretability and decision-making.

Sample Dataset

Student_ID	Study_Hours	Test_Score	Attendance (%)	Performance
S1	2	40	60	Poor
S2	3	50	65	Average
S3	4	55	70	Average
S4	5	65	75	Good
S5	6	75	80	Good
S6	7	85	85	Good
S7	8	90	88	Excellent
S8	9	95	90	Excellent
S9	10	98	92	Excellent
S10	11	100	95	Excellent

Tasks

A. Equal-Width Binning

i. Apply **3 bins** for Study_Hours and Test_Score
ii. Calculate bin width and define intervals

B. Equal-Frequency Binning

i. Divide data into **3 bins with equal records**
ii. Assign values accordingly

C. Concept Hierarchy Construction

- Study_Hours → Low / Medium / High

- Test_Score → Poor / Average / Excellent

4. Use the two methods below to normalize the following group of data:
200, 300,400, 600, 1000
(a) min-max normalization by setting min = 0 and max = 1
(b) z-score normalization

5. Data Transformation in Online Shopping Behavior Analysis *(L3, L4, L5)*

An **e-commerce company** is analyzing customer engagement by measuring the **time spent (in seconds)** on product pages. The duration varies significantly—from a few seconds to more than an hour—resulting in highly skewed data.

To improve the effectiveness of **customer segmentation using clustering techniques**, the dataset must be properly transformed and normalized.

D Sample Dataset: Visit Duration

Customer_ID	**Visit_Duration (seconds)**
C1	15
C2	30
C3	45
C4	60
C5	120
C6	300
C7	600
C8	900
C9	1800
C10	3600

✅ Tasks

A. Equal-Frequency Binning

i. Divide the dataset into **3 bins** with approximately equal number of records
ii. Assign each value to its respective bin
iii. Analyze how effectively this method groups short and long visit durations

B. Z-Score Normalization

i. Compute the **mean and standard deviation** of visit duration
ii. Apply Z-score normalization:

$$Z = \frac{X - \mu}{\sigma}$$

iii. Discuss how this method handles **extreme values (e.g., 1800, 3600 seconds)**

C. Min-Max Normalization

i. Apply Min-Max normalization using range **[0,1]**:

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

ii. Transform all values accordingly
iii. Compare results with Z-score normalization in terms of **scaling and sensitivity to outliers**

SOLUTIONS:

Q1. Data Preprocessing in Retail Analytics

Understanding of the Scenario:

A retail analytics company is developing a **recommendation system** to suggest products to customers based on their purchase history and shopping behavior. The customer data has been collected from multiple stores and through manual entries. Because of this, the dataset contains several data quality issues such as:

- Missing values
- Inconsistent categorical values
- Outliers (extreme values)

If these problems are not corrected before training the recommendation model, the predictions may become inaccurate and unreliable. Therefore, **data preprocessing** is an essential first step to clean and prepare the dataset.

Customer_ID	Age	Gender	Monthly_Spend (₹K)	Purchase_Frequency	Loyalty_Level
C1	23	M	20	5	Low
C2	NaN	Male	35	8	Medium
C3	31	F	NaN	10	High
C4	105	Female	500	50	High

Customer_ID	Age	Gender	Monthly_Spend (₹K)	Purchase_Frequency	Loyalty_Level
C5	27	male	25	NaN	Low
C6	40	FEMALE	45	12	Medium
C7	36	M	38	9	Low
C8	42	Female	300	45	High

A. Handling Missing Values
What are Missing Values?
Missing values occur when some information is not available in the dataset. They may happen because:
- Customers skipped some details.
- Data entry mistakes occurred.
- Technical problems happened while collecting data.
- Data from different stores was merged incorrectly.

Missing values reduce the quality of the dataset and may affect machine learning models.

Step 1: Identify Missing Values
From the dataset:

Column	Missing Record
Age	C2
Monthly_Spend	C3
Purchase_Frequency	C5

There are **three missing values**.

Step 2: Missing Value Imputation
Instead of deleting records, we estimate the missing values. This process is called **imputation**.

Method 1: Median Imputation
Why Median?
The dataset contains very high values such as:
- Age = **105**
- Monthly Spend = **500**

These outliers affect the mean but **do not affect the median much**. Therefore, median is more suitable.

Age Imputation
Available Age values:
23, 31, 105, 27, 40, 36, 42
Arrange in ascending order:
23, 27, 31, **36**, 40, 42, 105
Median = **36**
Therefore,
C2 Age = 36

Monthly Spend Imputation
Available values:
20, 35, 500, 25, 45, 38, 300
Sorted:
20, 25, 35, **38**, 45, 300, 500
Median = **38**
Therefore,
C3 Monthly Spend = 38

Purchase Frequency
Available values:
5, 8, 10, 50, 12, 9, 45
Sorted:
5, 8, 9, **10**, 12, 45, 50
Median = **10**

Therefore,
C5 Purchase Frequency = 10

Method 2: k-Nearest Neighbour (k-NN) Imputation
Concept
k-NN finds records similar to the missing record and uses the average of their values.
Example:
Customer C3 belongs to **High Loyalty**.
Similar customers:
- C4 → Spend = 500
- C8 → Spend = 300

Estimated Spend

$$= \frac{500 + 300}{2} = 400$$

Therefore,
Estimated Monthly Spend = **400**

Comparison

Median Imputation	k-NN Imputation
Simple	More complex
Faster	Slower
Robust to outliers	Can be affected by outliers
Best for this dataset	Better when similar records exist

Decision
Since the dataset contains extreme values, **Median Imputation** is the better choice because it is not influenced by outliers.

B. Outlier Detection and Treatment
What is an Outlier?
An outlier is a value that is **much larger or much smaller** than the rest of the data.
Examples:
- Age = 105
- Monthly Spend = 500

These values are very different from the other observations.
Outliers can reduce the accuracy of machine learning models.

Z-Score Method
Formula

$$Z = \frac{X - \mu}{\sigma}$$

Where:
- X = Data value
- μ = Mean
- σ = Standard deviation

Interpretation
- If **|Z| < 3**, the value is considered normal.
- If **|Z| > 3**, the value is considered an outlier.

Identify Outliers
Age
Most ages:
23–42 years
One customer:
105 years

This is much higher than the others and is treated as an outlier.

Monthly Spend
Normal values:
20–45
Extreme values:
300
500
These values are much larger than the remaining observations and are considered outliers.

Outlier Records

Customer	**Outlier**
C4	Age = 105, Spend = 500
C8	Spend = 300

Treatment of Outliers
There are several methods:
1. Removal
Delete the record completely.
Not recommended here because C4 and C8 may represent genuine high-value customers.

2. Capping (Winsorization)
Replace extreme values with reasonable limits.
Example:
Age
105 → 60
Monthly Spend
500 → 100
300 → 100

3. Transformation
Use logarithmic transformation.
Example:
Instead of using 500 directly,
Use

$$\log(500)$$

This reduces the effect of extreme values.

Decision
The best method is **Capping** because:

- Important customers are not removed.
- Extreme influence is reduced.
- The dataset remains balanced.

C. Categorical Data Standardization
Why Standardization?
Machine learning algorithms treat different spellings as different categories.
Example:
Male
male
M
These represent the same gender but are interpreted as different values.
This creates inconsistency.

Identify Inconsistencies

Male

- M
- Male
- male

Female

- F
- Female
- FEMALE

Standardized Dataset

Original	Standard
M	Male
Male	Male
male	Male
F	Female
Female	Female
FEMALE	Female

Benefits

- Removes duplicate categories.
- Improves model accuracy.
- Makes analysis easier.
- Reduces errors during encoding.

Final Cleaned Dataset

Customer	Age	Gender	Monthly Spend	Frequency	Loyalty
C1	23	Male	20	5	Low
C2	36	Male	35	8	Medium
C3	31	Female	38	10	High
C4	60	Female	100	50	High
C5	27	Male	25	10	Low
C6	40	Female	45	12	Medium
C7	36	Male	38	9	Low
C8	42	Female	100	45	High

Q2. Covariance & Data Reduction in E-commerce Analytics

Understanding of the Scenario:

An e-commerce platform wants to develop a **predictive model** to identify **high-value customers**. High-value customers are those who purchase products frequently, spend more money, and contribute significantly to the company's revenue.
The dataset contains customer information such as:

- Age
- Income
- Purchase Value
- Credit Score
- Number of Orders
- Average Balance
- EMI
- Customer Class

Many of these features are related to each other. For example, customers with higher income often have higher purchase values and higher average balances. Such similar features increase computational cost and may reduce model efficiency. Therefore, the company needs to perform **covariance analysis, feature selection, and data transformation** before building the predictive model.

Sample Dataset

Cust_ID	**Age**	**Income (₹K)**	**Purchase_Value (₹K)**	**Credit_Score**	**Orders**	**Avg_Balance (₹K)**	**EMI (₹)**	**Customer_Class**
U1	24	40	100	650	15	12	5000	Low
U2	34	60	150	700	20	18	7000	Medium
U3	44	85	210	720	28	25	9000	Medium
U4	52	120	320	800	35	40	15000	High
U5	29	45	110	680	18	14	5500	Low
U6	39	75	180	710	24	22	8500	Medium
U7	56	150	360	820	40	50	18000	High
U8	37	70	160	705	22	20	8000	Medium

A. Covariance Analysis
What is Covariance?
Covariance is a statistical measure used to determine **how two variables change together**.
It helps answer questions such as:

- Does Purchase Value increase when Income increases?
- Do customers with higher income spend more?
- Are two variables related?

Covariance is commonly used during **data preprocessing** to identify relationships between features.

Formula

$$Cov(X\text{,}Y) = \frac{\sum(X - \bar{X})(Y - \bar{Y})}{n - 1}$$

Where:

- **X** = Income
- **Y** = Purchase Value
- **X̄** = Mean of Income
- **Ȳ** = Mean of Purchase Value
- **n** = Number of observations

Step 1: Calculate Mean

Income

$$\bar{X} = \frac{40 + 60 + 85 + 120 + 45 + 75 + 150 + 70}{8}$$

$$= \frac{645}{8} = 80.625$$

Purchase Value

$$\bar{Y} = \frac{100 + 150 + 210 + 320 + 110 + 180 + 360 + 160}{8}$$

$$= \frac{1590}{8} = 198.75$$

Step 2: Covariance Calculation Table

Customer	Income (X)	Purchase (Y)	$X - \bar{X}$	$Y - \bar{Y}$	$(X - \bar{X})(Y - \bar{Y})$
U1	40	100	-40.625	-98.75	4011.72
U2	60	150	-20.625	-48.75	1005.47
U3	85	210	4.375	11.25	49.22
U4	120	320	39.375	121.25	4774.22
U5	45	110	-35.625	-88.75	3161.72
U6	75	180	-5.625	-18.75	105.47
U7	150	360	69.375	161.25	11187.89
U8	70	160	-10.625	-38.75	411.72

Sum of products = 14707.43 (approx.)

$$Cov(X, Y) = \frac{14707.43}{7}$$

$$\boxed{Cov(X, Y) \approx 2101.06}$$

(The exact value may vary slightly depending on rounding.)

Interpretation of Covariance

The covariance is **positive**, which means:

- When **Income increases**, **Purchase Value also increases**.
- Customers with higher salaries generally purchase more products.
- Customers with lower incomes generally spend less.

A positive covariance indicates that both variables move in the **same direction**.

Types of Covariance

1. Positive Covariance

Both variables increase or decrease together.

Example:

- Higher Income → Higher Purchase Value

2. Negative Covariance

One variable increases while the other decreases.

Example:

- Higher Loan Amount → Lower Savings

3. Zero Covariance

There is no relationship between the variables.

Why is Covariance Important?

Covariance helps:

- Understand relationships between variables.
- Detect redundant attributes.
- Improve feature selection.

- Reduce computational cost.
- Increase model accuracy.

B. Feature Selection
What is Feature Selection?
Feature selection is the process of selecting only the **most useful variables** for model building while removing irrelevant or redundant features.
Instead of using every column, only the important columns are selected.

Why is Feature Selection Needed?
Using too many features can:

- Increase training time.
- Increase memory usage.
- Cause overfitting.
- Reduce prediction accuracy.
- Make the model unnecessarily complex.

Identify Redundant Attributes
Income and Average Balance
Customers with higher income usually maintain higher average balances.
These two features provide similar information.

Income and EMI
Customers with higher incomes often have larger EMIs because they can afford bigger loans.
These features are related.

Purchase Value and Orders
Customers who place more orders usually have higher purchase values.
These variables are also highly correlated.

Selected Features
Keep:

- Age
- Income
- Purchase Value
- Credit Score
- Orders

Remove:

- Avg_Balance
- EMI

Justification
The selected features provide sufficient information about:

- Customer demographics (Age)
- Financial capability (Income)
- Spending behavior (Purchase Value)
- Creditworthiness (Credit Score)
- Shopping frequency (Orders)

Removing redundant features reduces complexity while maintaining prediction quality.

C. Data Transformation (Standardization)
What is Standardization?
Standardization converts numerical values into a common scale so that all features contribute equally during model training.
Without standardization:

- Income ranges from **40–150**
- EMI ranges from **5000–18000**

The EMI values are much larger and may dominate the learning process.

Formula

$$Z = \frac{X - \mu}{\sigma}$$

Where:

- X = Original value
- μ = Mean
- σ = Standard deviation

Example Calculation
Income = **40**
Mean = **80.625**
Standard deviation ≈ **39.6**

$$Z = \frac{40 - 80.625}{39.6}$$
$$= -1.03$$

This means the income of ₹40K is **1.03 standard deviations below the average income**.
The same formula is applied to all numerical attributes except **Cust_ID** and **Customer_Class**.

Why is Standardization Required?
Standardization:

- Brings all features to the same scale.
- Prevents large-valued features from dominating.
- Improves convergence during training.
- Makes distance calculations more accurate.
- Improves the performance of algorithms such as:
    - K-Nearest Neighbors (KNN)
    - Support Vector Machine (SVM)
    - Principal Component Analysis (PCA)
    - Neural Networks
    - Clustering algorithms

Decision Making
After performing the analysis:

- Covariance indicates a **positive relationship** between Income and Purchase Value.
- Redundant features (**Average Balance** and **EMI**) should be removed to simplify the model.
- Standardization should be applied to all numerical features before training the predictive model.

These preprocessing steps improve model efficiency, reduce computational cost, and increase prediction accuracy.

Conclusion
In this scenario, covariance analysis shows that customers with higher incomes tend to have higher purchase values. Feature selection removes redundant variables, reducing complexity and avoiding overfitting. Standardization ensures that all numerical features are on the same scale, improving the performance of machine learning algorithms. Together, these preprocessing techniques help the e-commerce company build a faster, more accurate, and more reliable model for identifying high-value customers.

Q3. Data Transformation & Discretization in Education Analytics

An education analytics team wants to build a machine learning model to predict **student performance**. The dataset contains continuous numerical values such as **Study Hours**, **Test Scores**, and **Attendance**. While continuous data is useful for calculations, it is sometimes difficult to interpret.
For example:

- A student studying **2 hours** and another studying **3 hours** may have similar learning levels.
- Instead of using exact values, it is often easier to group students into categories such as **Low, Medium, and High**.

This process is called **Discretization**.
Discretization converts continuous numerical data into a limited number of categories (bins), making the data easier to understand and analyze.

Sample Dataset

Student_ID	Study_Hours	Test_Score	Attendance (%)	Performance
S1	2	40	60	Poor
S2	3	50	65	Average
S3	4	55	70	Average
S4	5	65	75	Good
S5	6	75	80	Good
S6	7	85	85	Good
S7	8	90	88	Excellent
S8	9	95	90	Excellent
S9	10	98	92	Excellent
S10	11	100	95	Excellent

A. Equal-Width Binning
What is Equal-Width Binning?
Equal-width binning divides the data range into **equal-sized intervals**.
Each bin has the same width, but the number of records in each bin may differ.

Formula

$$\text{Bin Width} = \frac{\text{Maximum Value} - \text{Minimum Value}}{\text{Number of Bins}}$$

Step 1: Study Hours
Minimum = **2**
Maximum = **11**
Number of bins = **3**

$$\text{Width} = \frac{11 - 2}{3} = 3$$

Therefore, the intervals are:

- **Bin 1:** 2 – 5
- **Bin 2:** 5 – 8
- **Bin 3:** 8 – 11

Study Hours after Binning

Study Hours	Bin
2	Low
3	Low
4	Low
5	Medium

Study Hours	Bin
6	Medium
7	Medium
8	High
9	High
10	High
11	High

Step 2: Test Score
Minimum = **40**
Maximum = **100**
Number of bins = **3**

$$\text{Width} = \frac{100 - 40}{3} = 20$$

Intervals:

- **40–60**
- **60–80**
- **80–100**

Test Score after Binning

Score	Bin
40	Low
50	Low
55	Low
65	Medium
75	Medium
85	High
90	High
95	High
98	High
100	High

Advantages of Equal-Width Binning

- Simple to understand.
- Easy to implement.
- Useful when data is evenly distributed.

Disadvantages

- Some bins may contain many records while others contain very few.
- Sensitive to outliers.

B. Equal-Frequency Binning
What is Equal-Frequency Binning?
Equal-frequency binning divides the dataset so that **each bin contains approximately the same number of records**, regardless of the interval width.
Since there are **10 records** and **3 bins**:

$$10 \div 3 \approx 3\text{–}4 \text{ records per bin}$$

Study Hours
Sorted values:
2,3,4,5,6,7,8,9,10,11

Bins

Bin 1
2,3,4

Bin 2
5,6,7

Bin 3
8,9,10,11

Table

Study Hours	Equal Frequency Bin
2	Low
3	Low
4	Low
5	Medium
6	Medium
7	Medium
8	High
9	High
10	High
11	High

Test Scores
Sorted:
40,50,55,65,75,85,90,95,98,100

Bins
Bin 1
40,50,55

Bin 2
65,75,85

Bin 3
90,95,98,100

Table

Score	Equal Frequency Bin
40	Low
50	Low
55	Low
65	Medium
75	Medium
85	Medium
90	High
95	High
98	High
100	High

Advantages

- Every bin contains nearly the same number of records.
- Reduces data imbalance.
- Suitable for skewed datasets.

Disadvantages

- Bin widths are not equal.
- Interpretation of intervals is less intuitive.

C. Concept Hierarchy Construction
What is a Concept Hierarchy?
A concept hierarchy converts **detailed numerical values** into **higher-level categories**.
Instead of exact values, broader categories are used.
This makes the data easier to understand and analyze.

Study Hours Hierarchy

Study Hours	Category
2–4	Low
5–7	Medium
8–11	High

Test Score Hierarchy

Test Score	Category
40–60	Poor
61–80	Average
81–100	Excellent

Hierarchy Diagram
Study Hours

Low Medium High
Test Score

Poor Average Excellent

Why Concept Hierarchy is Useful?

- Simplifies complex numerical data.
- Improves readability.
- Helps teachers identify student performance groups.
- Useful in decision trees and rule-based systems.
- Supports educational decision-making.

Comparison of Equal-Width and Equal-Frequency Binning

Feature	Equal-Width	Equal-Frequency
Bin Size	Equal interval width	Equal number of records
Distribution	May be uneven	Balanced
Sensitive to Outliers	Yes	Less sensitive
Easy to Interpret	Yes	Moderate
Suitable For	Uniform data	Skewed data

Decision Making
Based on the analysis:

- **Equal-width binning** is simple and suitable when data is evenly distributed.
- **Equal-frequency binning** ensures that each bin has nearly the same number of students, making it suitable for uneven datasets.
- **Concept hierarchy** further improves interpretation by grouping students into meaningful categories such as Low, Medium, High, Poor, Average, and Excellent.

These techniques make the dataset easier to analyze and improve the performance and interpretability of predictive models.

Conclusion
Data transformation and discretization are important preprocessing techniques in education analytics. Equal-width binning groups values into equal intervals, while equal-frequency binning ensures balanced groups with approximately the same number of records. Concept hierarchies simplify continuous numerical values into meaningful categories, making it easier for teachers, administrators, and machine learning models to analyze student performance and make informed decisions.
Understanding of the Scenario:

Q4. Normalization Techniques

Understanding of the Scenario:

In data mining and machine learning, datasets often contain attributes with different ranges of values. For example, one attribute may range from **200 to 1000**, while another may range from **0 to 10**. If these attributes are used directly, the larger values can dominate the smaller ones, affecting the performance of machine learning algorithms.
To solve this problem, **normalization** is applied. Normalization transforms data into a common scale without changing the relationship between the values.
In this question, the given data is:
200, 300, 400, 600, 1000
We need to normalize the data using:

1. **Min-Max Normalization**
2. **Z-Score Normalization**

What is Normalization?
Normalization is a **data preprocessing technique** used to scale numerical values into a common range.
Why is Normalization Needed?

- Eliminates differences in data scales.
- Improves the accuracy of machine learning models.
- Prevents large values from dominating small values.
- Makes calculations easier.
- Speeds up model training.

Given Data

Observation	Value
X_1	200
X_2	300
X_3	400
X_4	600
X_5	1000

A. Min-Max Normalization
What is Min-Max Normalization?
Min-Max Normalization transforms data into a specified range, usually **0 to 1**.
It preserves the relative relationship between the data values.

Formula

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Where:

- X= Original value
- X_{min}= Minimum value
- X_{max}= Maximum value

- X'= Normalized value

Step 1: Find Minimum and Maximum
Minimum value

$$X_{min} = 200$$

Maximum value

$$X_{max} = 1000$$

Range

$$1000 - 200 = 800$$

Step 2: Calculate Normalized Values
For 200

$$= \frac{200 - 200}{800} = \frac{0}{800} = 0$$

For 300

$$= \frac{300 - 200}{800} = \frac{100}{800} = 0.125$$

For 400

$$= \frac{400 - 200}{800} = \frac{200}{800} = 0.25$$

For 600

$$= \frac{600 - 200}{800} = \frac{400}{800} = 0.50$$

For 1000

$$= \frac{1000 - 200}{800} = \frac{800}{800} = 1$$

Min-Max Normalized Table

Original Value	Normalized Value
200	0.000
300	0.125
400	0.250
600	0.500
1000	1.000

Interpretation
- The minimum value (200) becomes **0**.
- The maximum value (1000) becomes **1**.
- All other values lie between **0 and 1**.
- The order of the data remains unchanged.

Advantages of Min-Max Normalization
- Very simple to calculate.
- Produces values within a fixed range (0–1).
- Useful for Neural Networks and Deep Learning.
- Preserves the relationship between values.

Disadvantages

- Highly sensitive to outliers.
- If a new maximum value appears later, normalization must be recalculated.

B. Z-Score Normalization

What is Z-Score Normalization?

Z-Score Normalization transforms data so that:

- Mean becomes **0**
- Standard deviation becomes **1**

Instead of limiting values between 0 and 1, it measures how far each value is from the average.

Formula

$$Z = \frac{X - \mu}{\sigma}$$

Where:

- X= Original value
- μ= Mean
- σ= Standard deviation

Step 1: Calculate Mean

$$\mu = \frac{200 + 300 + 400 + 600 + 1000}{5}$$
$$= \frac{2500}{5} = 500$$

Mean = **500**

Step 2: Calculate Standard Deviation

X	X − μ	(X − μ)²
200	-300	90000
300	-200	40000
400	-100	10000
600	100	10000
1000	500	250000

$$\sum(X - \mu)^2 = 400000$$

Population variance

$$= \frac{400000}{5} = 80000$$

Standard deviation

$$\sigma = \sqrt{80000}$$
$$\sigma \approx 282.84$$

Step 3: Calculate Z-Scores

For 200

$$Z = \frac{200 - 500}{282.84} = -1.06$$

For 300

$$= \frac{-200}{282.84} = -0.71$$

For 400

$$= \frac{-100}{282.84} = -0.35$$

For 600

$$= \frac{100}{282.84} = 0.35$$

For 1000

$$= \frac{500}{282.84} = 1.77$$

Z-Score Normalization Table

Original Value	Z-Score
200	-1.06
300	-0.71
400	-0.35
600	0.35
1000	1.77

Interpretation

- Negative values indicate observations below the mean.
- Positive values indicate observations above the mean.
- The value **1000** has the highest positive Z-score because it is farthest above the mean.

Advantages of Z-Score Normalization

- Less affected by changes in data range.
- Suitable when data follows a normal distribution.
- Commonly used in statistical analysis.
- Works well with algorithms like SVM, Logistic Regression, and PCA.

Disadvantages

- Does not scale data into a fixed range.
- Requires calculation of mean and standard deviation.

Comparison of Min-Max and Z-Score Normalization

Feature	Min-Max Normalization	Z-Score Normalization
Formula	$(X - X_{min})/(X_{max} - X_{min})$	$(X - \mu)/\sigma$
Output Range	0 to 1	No fixed range
Uses Mean & SD	No	Yes
Sensitive to Outliers	Yes	Less sensitive
Best Used For	Neural Networks, Deep Learning	SVM, PCA, Statistical Models

Decision Making

Based on the analysis:

- **Min-Max Normalization** is suitable when all values need to be scaled between **0 and 1**.
- **Z-Score Normalization** is preferred when data follows a normal distribution or when algorithms require standardized data.

For this dataset, both methods correctly normalize the values, but the choice depends on the machine learning algorithm being used.

Conclusion
Normalization is an important data preprocessing technique that improves the quality of machine learning models by bringing all numerical values to a common scale. In this question, Min-Max Normalization converted the values into the range **0 to 1**, while Z-Score Normalization transformed the values based on the mean and standard deviation. Both methods help improve model accuracy, reduce bias caused by different scales, and make the data more suitable for analysis.

Q5. Data Transformation in Online Shopping Behavior Analysis

Understanding of the Scenario:
An e-commerce company wants to analyze **customer engagement** by measuring the **time spent (in seconds)** on product pages.
The dataset contains visit durations ranging from **15 seconds to 3600 seconds (1 hour)**. Since the values vary significantly, the data is **highly skewed**.
If this raw data is directly used for clustering algorithms such as **K-Means**, customers with very large visit durations may dominate the results. Therefore, the data must be transformed using **Equal-Frequency Binning, Z-Score Normalization, and Min-Max Normalization** before analysis.
These preprocessing techniques improve the quality of the dataset and make clustering more effective.

Sample Dataset

Customer_ID	Visit Duration (seconds)
C1	15
C2	30
C3	45
C4	60
C5	120
C6	300
C7	600
C8	900
C9	1800
C10	3600

A. Equal-Frequency Binning
What is Equal-Frequency Binning?
Equal-frequency binning is a discretization technique in which the dataset is divided into bins containing **approximately the same number of records**.
Unlike equal-width binning, the intervals are not equal. Instead, each bin has almost the same number of observations.

Step 1: Arrange the Data
The data is already sorted.
15, 30, 45, 60, 120, 300, 600, 900, 1800, 3600
Total records = **10**
Number of bins = **3**
Each bin should contain approximately

$$\frac{10}{3} \approx 3\text{–}4 \text{ records}$$

Step 2: Create Bins
Bin 1 (Short Visits)
15
30
45
60

Bin 2 (Medium Visits)
120
300
600

Bin 3 (Long Visits)
900
1800
3600

Table

Customer	Visit Duration	Bin
C1	15	Short
C2	30	Short
C3	45	Short
C4	60	Short
C5	120	Medium
C6	300	Medium
C7	600	Medium
C8	900	Long
C9	1800	Long
C10	3600	Long

Analysis
The customers are grouped into three categories:
- **Short Visits** → Less engagement.
- **Medium Visits** → Moderate engagement.
- **Long Visits** → High engagement.

Equal-frequency binning ensures that each category contains almost the same number of customers, making comparisons fairer.

Advantages
- Balances the number of records in each group.
- Useful for skewed datasets.
- Improves clustering performance.

Disadvantages
- Interval widths are unequal.
- Less intuitive than equal-width binning.

B. Z-Score Normalization
What is Z-Score Normalization?
Z-Score Normalization transforms the data based on the **mean** and **standard deviation**.
Instead of restricting values between 0 and 1, it measures how far each value is from the average.

Formula

$$Z = \frac{X - \mu}{\sigma}$$

Where:
- X = Original value
- μ = Mean
- σ = Standard deviation

Step 1: Calculate Mean

$$\mu = \frac{15 + 30 + 45 + 60 + 120 + 300 + 600 + 900 + 1800 + 3600}{10}$$

$$= \frac{7470}{10} = 747$$

Mean = **747 seconds**

Step 2: Calculate Standard Deviation
Using the standard deviation formula,

$$\sigma \approx 1108.5$$

Step 3: Calculate Z-Scores
For 15

$$Z = \frac{15 - 747}{1108.5} = -0.66$$

For 30

$$= -0.65$$

For 45

$$= -0.63$$

For 60

$$= -0.62$$

For 120

$$= -0.56$$

For 300

$$= -0.40$$

For 600

$$= -0.13$$

For 900

$$= 0.14$$

For 1800

$$= 0.95$$

For 3600

$$= 2.57$$

Z-Score Table

Visit Duration	Z-Score
15	-0.66
30	-0.65
45	-0.63
60	-0.62

Visit Duration	Z-Score
120	-0.56
300	-0.40
600	-0.13
900	0.14
1800	0.95
3600	2.57

Interpretation

- Negative Z-scores indicate visit durations **below the average**.
- Positive Z-scores indicate durations **above the average**.
- **3600 seconds** has the highest Z-score, showing it is an extreme value compared to the average.

How Does Z-Score Handle Extreme Values?
Z-score does **not remove** extreme values. Instead, it expresses how far they are from the mean.
For example:

- 3600 seconds → Z = 2.57

This indicates the value is **2.57 standard deviations above the mean**.
This helps in identifying unusual customer behavior.

C. Min-Max Normalization
What is Min-Max Normalization?
Min-Max Normalization scales all values into a fixed range, usually **0 to 1**.

Formula

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Step 1: Find Minimum and Maximum
Minimum = **15**
Maximum = **3600**
Range

$$3600 - 15 = 3585$$

Step 2: Normalize All Values
15

$$= \frac{15 - 15}{3585} = 0$$

30

$$= \frac{15}{3585} = 0.004$$

45

$$= \frac{30}{3585} = 0.008$$

60

$$= \frac{45}{3585} = 0.013$$

120

$$= \frac{105}{3585} = 0.029$$

300

$$= \frac{285}{3585} = 0.079$$

600

$$= \frac{585}{3585} = 0.163$$

900

$$= \frac{885}{3585} = 0.247$$

1800

$$= \frac{1785}{3585} = 0.498$$

3600

$$= \frac{3585}{3585} = 1$$

Min-Max Table

Visit Duration	Normalized Value
15	0.000
30	0.004
45	0.008
60	0.013
120	0.029
300	0.079
600	0.163
900	0.247
1800	0.498
3600	1.000

Comparison of Z-Score and Min-Max Normalization

Feature	Z-Score	Min-Max
Formula	$(X - \mu)/\sigma$	$(X - X_{min})/(X_{max} - X_{min})$
Range	No fixed range	0–1
Uses Mean & SD	Yes	No
Sensitive to Outliers	Less	More
Best For	Statistical models, SVM, PCA	Neural Networks, Clustering

Interpretation

- **Min-Max Normalization** scales all values between **0 and 1**, making them suitable for algorithms that require fixed ranges.

- **Z-Score Normalization** measures how far each value is from the mean and is better for identifying extreme values.

Decision Making
Based on the analysis:
- Equal-frequency binning effectively groups customers into **short, medium, and long visit duration** categories.
- Z-score normalization identifies customers with unusually long visit durations.
- Min-Max normalization scales the data into the range **0–1**, making it suitable for clustering algorithms such as K-Means.

Conclusion
Data transformation is an essential preprocessing step in online shopping analytics. Equal-frequency binning simplifies customer engagement into meaningful groups. Z-score normalization standardizes the data and helps identify extreme visit durations, while Min-Max normalization scales all values into a common range. Together, these techniques improve clustering performance, enhance customer segmentation, and enable the company to better understand customer behavior for personalized marketing and recommendation systems.